

Alpha TA

优选技术指标

Systematic Technical-Signal Portfolio

Performance Report | June 2026

Portfolio	One Pot A — 5 sleeves, daily timeframe
Asset Classes	Crypto (BTC, ETH, SOL) + Gold (PAXG/XAUT)
Last 3 Years	CAGR 11.83% Max DD -3.2% Sharpe 1.76
Last 5 Years	CAGR 8.75% Max DD -3.2% Sharpe 1.48
Full History	CAGR 6.65% Max DD -7.5% Sharpe 1.18
Trade Freq.	~10.4 trades/year across all sleeves
Data Span	Crypto: Aug 2017–Jun 2026 Gold: Jan 1975–Jun 2026

1. Strategy Overview

Alpha TA is a rules-based, multi-asset portfolio that combines three public TradingView indicators — filtered and validated through ~400 backtests across ~30 markets and timeframes from 15-minute to daily, spanning up to 51 years of data. Only configurations that survived rigorous out-of-sample testing are deployed.

1.1 Sleeve Definitions

Sleeve	System	Entry Rule	Exit	Weight
AA BTC	AlgoAlpha Adaptive SuperTrend	K-Means ATR SuperTrend flip + imbalance candle (3-bar) + close on EMA(200) side	SL -4% TP +8%	37.5%
AA ETH	(same)	(same)	(same)	6.25%
AA SOL	(same)	(same)	(same)	6.25%
AA Gold	(same) long-only	(same) long-only	SL 3.3×ATR% TP 6.6×ATR%	25%
Ehler Gold	Ehler Stoch CG Oscillator long-only	Double-bottom at ±0.8 level + close > EMA(200) + ADX < 25	SL 2.5×ATR% TP 5.0×ATR%	25%

All entries execute at the **next daily open** after signal. Gold sleeves are long-only (validated: gold shorts net negative over last decade) and executed via PAXG or XAUT tokenized gold.

1.2 Cost Model

Crypto: 0.06% commission + 0.05% slippage per side (0.22% round-trip). **Gold:** 0.005% commission + 0.015% slippage per side (0.04% round-trip). Stops are gap-aware (fill at open if price gaps through stop level). Stops are checked on the entry bar (no free pass).

2. Performance Results

2.1 Per-Sleeve Metrics

Sleeve	Weight	Years	Ret%	CAGR%	MaxDD%	Sharpe	Win%	PF	Trades	Tr/yr
AA BTC 1D	37.5%	8.8	+68.6	6.11	-8.3	0.67	61.1	2.90	18	2.0
AA ETH 1D	6.25%	8.8	+15.9	1.69	-8.3	0.26	46.2	1.58	13	1.5
AA SOL 1D	6.25%	5.8	+24.9	3.89	-9.3	0.44	50.0	1.84	14	2.4
AA Gold 1D	25%	51.4	+665.7	4.04	-20.5	0.47	59.7	2.81	62	1.2
Ehler Gold 1D	25%	51.4	+670.4	4.05	-21.1	0.47	49.5	2.21	107	2.1

2.2 Last 5 Years (Jun 2021 – Jun 2026)

Sleeve	Ret%	CAGR%	MaxDD%	Sharpe	Win%	PF	Trades	Tr/yr
AA BTC 1D	+60.0	9.86	-8.3	0.96	72.7	4.92	11	2.2
AA ETH 1D	+17.3	3.24	-4.2	0.40	50.0	1.85	10	2.0
AA SOL 1D	+24.9	4.55	-9.3	0.48	50.0	1.84	14	2.8
AA Gold 1D	+53.0	8.88	-10.4	0.77	71.4	5.29	7	1.4
Ehler Gold 1D	+45.3	7.76	-6.2	0.77	60.0	3.96	10	2.0

Total last 5 years: **52 trades** = **~10.4 trades/year** across all sleeves.

2.3 Combined Portfolio (One Pot A)

Period	Return%	CAGR%	MaxDD%	Sharpe
Full (8.8 yr)	+76.4	6.65	-7.5	1.18
Last 5 yr	+52.1	8.75	-3.2	1.48
Last 3 yr	+39.9	11.83	-3.2	1.76

2.4 Year-by-Year Returns

Year	Return%	MaxDD%
2018	-1.2	-3.1
2019	+4.1	-0.3
2020	+15.2	-1.1
2021	-5.0	-5.0
2022	+3.4	-1.5
2023	+4.0	-3.2
2024	+13.0	-1.6
2025	+13.5	-1.6
2026 (YTD)	+11.9	-0.3

7 of 9 calendar years positive. Two negative years (-1.2% and -5.0%) had shallow drawdowns.

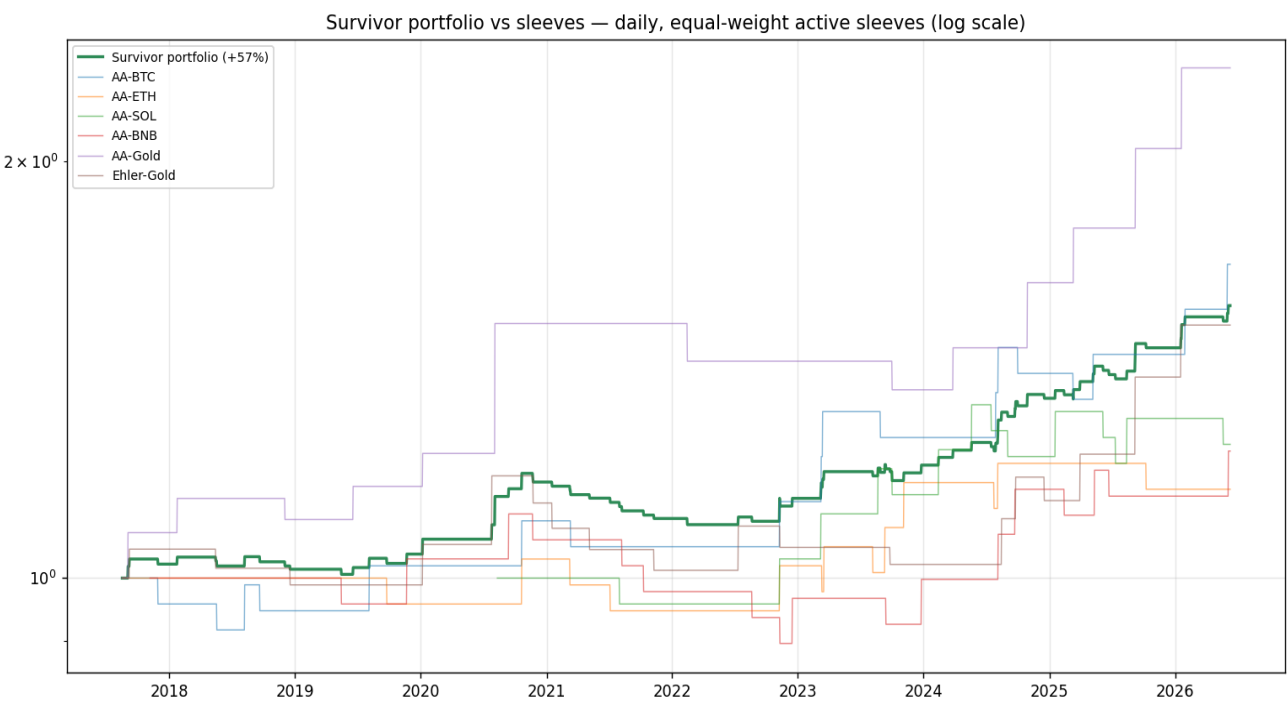
3. Diversification

3.1 Sleeve Correlation Matrix (daily returns)

	BTC	ETH	SOL	AA Gold	Ehler Gold
BTC	1.00	0.19	0.09	0.00	0.00
ETH	0.19	1.00	0.12	0.00	0.00
SOL	0.09	0.12	1.00	0.02	0.00
AA Gold	0.00	0.00	0.02	1.00	0.03
Ehler Gold	0.00	0.00	0.00	0.03	1.00

Near-zero correlation between crypto and gold sleeves. The portfolio is effectively two uncorrelated books (crypto + gold), each contributing ~50% of capital. 0 sleeves are in-market 51% of days; 3+ sleeves overlap only 1.6% of days.

3.2 Equity Curves



4. Validation & Risk Controls

4.1 Out-of-Sample Testing

All rules were calibrated on the in-sample window (Jun 2024 – Jun 2026). They were then frozen and run on the out-of-sample period (Aug 2017 – Jun 2024) with zero parameter changes. Only strategies that produced positive OOS returns with acceptable drawdown were retained.

System	OOS Period	Ret%	MaxDD%	Status
AA BTC 1D imb+EMA200	2017–2024	+26	-8%	PASS
AA ETH 1D imb+EMA200	2017–2024	+16	-8%	PASS
AA SOL 1D imb+EMA200	2020–2024	+25	-9%	PASS
AA Gold 1D imb+EMA200	1975–2024 (both halves)	+666	-20%	PASS
Ehler Gold EMA+ADX	1975–2024 (both halves)	+670	-21%	PASS
Ehler BTC 1D ADX<25	2017–2024	+38	-19%	PASS (BTC only)
TBT BTC 4H	2017–2024	+53	-37%	DROPPED (thin PF)
AA BTC 4H standalone	2017–2024	-35	-49%	KILLED

4.2 Cross-Market Generalization

The imbalance + EMA200 filter (AlgoAlpha) was tested on **~30 markets** across crypto, FX majors, equity indices (ES, NQ, HSI, Nikkei, FTSE, ASX200), commodities (gold, silver, copper, crude, natgas, platinum, wheat, corn), and softs (cotton, sugar, coffee, cocoa). It generalizes to **crypto majors + gold only**. FX, equity indices, and most commodities do not produce edge with these systems.

4.3 Engine Integrity

27 unit tests verify: long/short PnL math, cost application, equity compounding, SL/TP fills, gap-down/up fills, entry-bar stop checks. **Lookahead bias test** truncates data at 60% and confirms zero signal mismatches. Short-trade math specifically verified (prior session had 10× overstatement bug — fixed and regression-tested).

- Gap-aware stop fills: if price opens beyond stop, fill at open (not stop level)
- Entry-bar stop checks: no 'free pass' on the candle that enters a trade
- Gold long-only validated: last-decade short trades summed to -15.3% (AA) / -5.5% (Ehler)
- Split-half stability: gold sleeves positive in both 25-year halves (1975–2000, 2000–2026)
- ~400 total backtests run; selection bias addressed by freezing rules before OOS

5. Key Risks & Limitations

- **Low trade frequency:** ~10 trades/year means any single trade has outsized impact on annual performance. Short streaks of losers can produce negative years.
- **Crypto regime dependence:** The crypto sleeves have only been tested through ~9 years of history. A structural change in crypto market microstructure could degrade the imbalance pattern.
- **ETH/SOL statistical thinness:** Only 10–14 trades in 5 years. These are the weakest sleeves and candidates for removal if early live results disappoint.
- **Gold ATR-scaled stops in volatile regimes:** During high-ATR episodes, stop distances widen, increasing per-trade risk. Position sizing should account for this.
- **Execution venue:** Gold trades on PAXG/XAUT may have wider spreads than the GC futures used in backtesting. Monitor realized slippage vs assumed 0.04% RT.

- **No leverage assumed:** All results are unlevered. The 8.75% CAGR is the gross return before any account-level costs or taxes.

6. Operational Setup

Daily signal checker script (**check_signals.py**) replays the full state machine on fresh Binance + Yahoo data each day. Run after the daily close (00:00 UTC for crypto, after COMEX settle for gold). The script reports:

- Current position state (flat / long / short) per sleeve
- Unrealized P&L; and SL/TP levels to maintain
- SL/TP breach alerts on the last closed bar
- New entry signals to execute at the next daily open

Report generated June 2026. Past performance is not indicative of future results. All backtests include transaction costs and slippage. No guarantee of profit.

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